IDENTIFICATION OF THE MINIMUM INERTIAL PARAMETERS OF ROBOTS

M. GAUTIER and W. KHALIL

Laboratoire d'Automatique de Nantes URA C.N.R.S 256
1 Rue de la Noë
E.N.S.M. 44072 NANTES CEDEX
FRANCE

ABSTRACT

This paper presents a direct method to determine the set of minimum number of inertial parameters of tree structure robots. The given method permits to classify most of the regrouped parameters by means of closed form relations. The paper presents also a linear model for the identification of the minimum inertial parameters. The identification model is computationally simple, it is function of the joint positions and velocities and does not need to calculate the joint accelerations.

1- Introduction

In the last years many control schemes based on the dynamic model have been presented [1],...,[4]. Two problems have to be solved for on line implementation of these control laws:

1- The numerical values of the inertial parameters must be known. The solution of this problem has been investigated by the use of identification procedures based on a dynamic model linear in the inertial parameters [5],...,[9]. The classification of the identifiable parameters, the set of minimum number of inertial parameters, increases the robustness of the identification process [8].
2- The computational cost of the dynamic model must be reduced. The solution of this problem has been carried out by the use of the customized Newton-Euler method, and the use of the set of minimum inertial parameters[10],...,[13].

The definition and the determination of the minimum number of inertial parameters have been investigated on a case by case basis in [12, ..., 15]. In [16] we have presented a general and direct method to calculate the minimum inertial parameters of open loop robots, at the same time similar results concerning the special case of rotational robots whose successive axes are perpendicular or parallel have been given by Mayeda and al.[17]. In this paper we extend our work of minimum inertial parameters and identification [16,18] to the case of tree structure robots.

2- Description of the robot

The system to be considered is a tree structure mechanism Figure 1. The description of the system will be carried out by the modified Denavit and Hartenberg notation [19,20].Thanks to the use of this method the closed form relations of regrouping the inertial parameters has been obtained. The system has n joints and (n+1) links, link 0 is the base, while link n is an end effector. A coordinate frame j is assigned fixed with respect to link j. The $\mathbf{z}_j$ axis is along the axis of joint j, the $\mathbf{x}_j$ axis is along the common perpendicular to $\mathbf{z}_j$ and one of the succeeding joints axes on the same link.

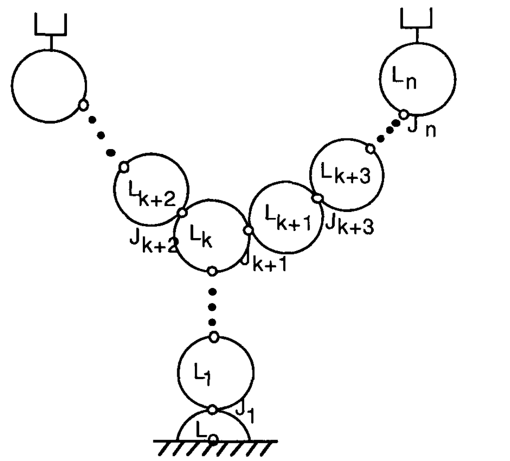

Figure 1

The frame j coordinate will be defined with respect to frame i, where i = a(j) indicate the link antecedent to link j, by the ${}^i\mathbf{T}_j$ matrix . Two cases are to be considered :
i- if $\mathbf{x}_i$ is perpendicular to $\mathbf{z}_i$ and $\mathbf{z}_j$ then ${}^i\mathbf{T}_j$ will be defined as, see Figure 2:

$$ {}^i\mathbf{T}_j = \begin{bmatrix} {}^i\mathbf{A}_j & {}^i\mathbf{P}_j \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} C\theta_j & -S\theta_j & 0 & d_j \\ C\alpha_j S\theta_j & C\alpha_j C\theta_j & -S\alpha_j & -r_j S\alpha_j \\ S\alpha_j S\theta_j & S\alpha_j C\theta_j & C\alpha_j & r_j C\alpha_j \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1) $$

where: ${}^i\mathbf{A}_j$ defines the orientation of frame j with respect to frame i and ${}^i\mathbf{P}_j$ defines the position of the origin of frame j with respect to frame i.
We note that if link i has at most two links or if the robot is of serial type we have only the foregoing case .

ii) if $\mathbf{x}_i$ is the common perpendicular to $\mathbf{z}_i$ and another joint axis as $\mathbf{z}_k$, Figure 3, in this case an auxiliary frame i' will be defined such that $\mathbf{z}_i'$ is along $\mathbf{z}_i$ while $\mathbf{x}_i'$ is along the common perpendicular to $\mathbf{z}_i$ and $\mathbf{z}_j$. Thus we have:

$$ {}^i\mathbf{T}_j = {}^i\mathbf{T}_{i'}\ {}^{i'}\mathbf{T}_j \quad (2) $$

where ${}^{i'}\mathbf{T}_j$ is the same as (1), while ${}^i\mathbf{T}_{i'}$ is equal to

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$$
{}^{i}\mathbf{T}_{i'} = \begin{bmatrix} C\gamma_j & -S\gamma_j & 0 & 0 \\ S\gamma_j & C\gamma_j & 0 & 0 \\ 0 & 0 & 1 & \varepsilon_j \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}
$$

The meaning of the parameters ; $\varepsilon_j, \gamma_j, \alpha_j, d_j, \theta_j, r_j$ can be found in [19,20] and can be deduced from Figures 2 and 3. The joint variable j will be denoted q_j such that:

$$
q_j = \bar{\sigma}_j \,\theta_j + \sigma_j r_j \tag{4}
$$

where: $\sigma_j = 0$ if joint j is rotational , $\sigma_j = 1$ for j translational, and $\bar{\sigma}_j = (1-\sigma_j)$.

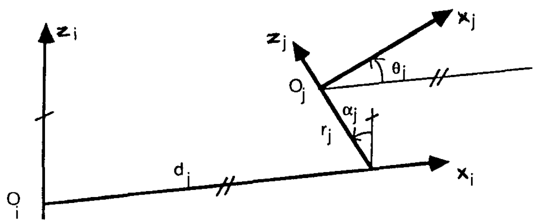

Figure 2

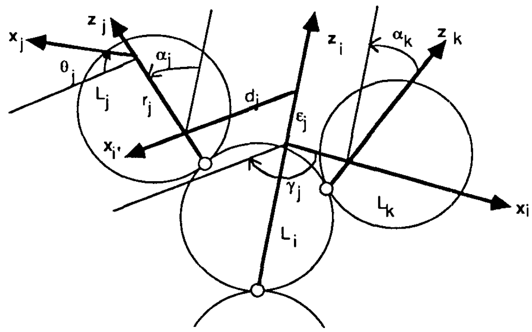

Figure 3

3- Minimum Inertial parameters

The minimum inertial parameters of the dynamic model can be obtained from the classical parameters by eliminating those which have no effect on the dynamic model and those which can be regrouped to some other parameters.

3-1 Classical inertial parameters

The classical inertial parameters are composed of the elements of:

- ${}^{j}\mathbf{J}_j$ the inertia matrix of link j about the origin of frame j,referred to frame j,
- ${}^{j}\mathbf{MS}_j$ the first moments of link j about the origin of frame j , referred to frame j,
- M_j the mass of link j.

Let:

$$
{}^{j}\mathbf{J}_j = \begin{bmatrix} XX_j & XY_j & XZ_j \\ XY_j & YY_j & YZ_j \\ XZ_j & YZ_j & ZZ_j \end{bmatrix}
$$

$$
{}^{j}\mathbf{MS}_j = [\, MXj \;\; MYj \;\; MZj]^T
$$

So the classical inertial parameters of link j will be represented by the vector $\mathbf{X}^j$, where:

$$
\mathbf{X}^j = [XX_j \;\; XY_j \;\; XZ_j \;\; YY_j \;\; YZ_j \;\; ZZ_j \;\; MX_j \;\; MY_j \;\; MZ_j \;\; M_j]^T
$$

The inertial parameters of the robot will be represented by the vector $\mathbf{X}$ containing the inertial parameters of all the links. Thus, in general the dimension of $\mathbf{X}$, denoted by m, is equal to 10 n.

3-2 The lagrangian dynamic model of a robot

The Lagrangian equation is given as:

$$
\Gamma_j = \frac{d}{dt}\left(\frac{\partial E}{\partial \dot{q}_j}\right) - \frac{\partial E}{\partial q_j} + \frac{\partial U}{\partial q_j} \qquad j = 1, \dots , n \tag{5}
$$

where: Γ_j is the torque (or force) of motor j,

E is the kinetic energy, and U is the potential energy of the robot. They are calculated as [13, 20]:

$$
E = \sum_{j=1}^{n} \frac{1}{2}[\, {}^{j}\omega_j^T\, {}^{j}\mathbf{J}_j\, {}^{j}\omega_j + M_j\, {}^{j}\mathbf{V}_j^T\, {}^{j}\mathbf{V}_j + 2\, {}^{j}\mathbf{MS}_j^T ({}^{j}\mathbf{V}_j \times {}^{j}\omega_j)] \tag{6}
$$

$$
U = -\sum_{j=1}^{n} [\, \mathbf{g}^T (M_j\, {}^{0}\mathbf{P}_j + {}^{0}\mathbf{A}_j\, {}^{j}\mathbf{MS}_j)] \tag{7}
$$

where:

${}^{j}\mathbf{V}_j$ the velocity of the origin of the link j fixed frame, referred to frame j,

${}^{j}\omega_j$ the angular velocity of link j, referred to frame j,

$\mathbf{g}$ is the acceleration of gravity.

${}^{j}\omega_j$ and ${}^{j}\mathbf{V}_j$ can be calculated by the following recursive equations [13, 20]:

$$
{}^{j}\omega_j = {}^{j}\mathbf{A}_i\, {}^{i}\omega_i + \bar{\sigma}_j\, \mathbf{Z}_0\, \dot{q}_j \tag{8}
$$

$$
{}^{j}\mathbf{V}_j = {}^{j}\mathbf{A}_i\, {}^{i}\mathbf{V}_i + {}^{j}\mathbf{A}_i\, ({}^{i}\omega_i \times {}^{i}\mathbf{P}_j) + \sigma_j\, \mathbf{Z}_0\, \dot{q}_j \tag{9}
$$

where $\mathbf{Z}_0 = [\,0 \quad 0 \quad 1]^T$, and i = a(j) indicates the link antecedent to link j.

From equations (6,...,9) we deduce that E and U are linear in the inertial parameters. Thus we can write:

$$
E = \sum_{i=1}^{m} \frac{\partial E}{\partial X_i} X_i = \sum_{i=1}^{m} DE_i\, X_i \tag{10}
$$

$$
U = \sum_{i=1}^{m} \frac{\partial U}{\partial X_i} X_i = \sum_{i=1}^{m} (dU_i + W_i)\, X_i = \sum_{i=1}^{m} DU_i\, X_i \tag{11}
$$

where X_i is an inertial parameter, and W_i is constant.

Using (5), (10) and (11) we obtain that the dynamic model is also linear in the inertial parameters. It can be given as:

$$
\Gamma = \mathbf{H}\;\mathbf{X} \tag{12}
$$

where: $\mathbf{H}$ is a nxm matrix, function of $\mathbf{q}$, $\dot{\mathbf{q}}$, $\ddot{\mathbf{q}}$ and the constant geometric parameters,

3-3 Determination of the parameters having no effect on the dynamic model

From (5), (10), (11), we see that a parameter X_i has no effect on Γ, if:

$$DE_i = 0 \text{ and } dU_i = 0 \tag{13}$$

Thus X_i is not an element of the minimum inertial parameters. This case concern the parameters of the first links, supposing that only one chain exists, to simplify the writting, and that r_1 is the first rotational joint and r_2 the first succeeding rotational joint which is not parallel to r_1, we can derive the following:

i- If joint j is translational and $j < r_1$ then XX_j, XY_j, XZ_j, YY_j, YZ_j, ZZ_j, MX_j, MY_j, MZ_j have no effect on the dynamic model, this due to the fact that $\omega_j = \mathbf{0}$ and ${}^0\mathbf{A}_j$ is constant.

ii- If all of the joints $j < r_1$ are parallel to the joint r_1, and to the direction of the acceleration of the gravity, or if $a(r_1) = 0$ and r_1 is parallel to the direction of gravity, then MX_{r1}, MY_{r1} have no effect on the dynamic model. Moreover if $a(r_1) = 0$ then M_{r1} has no effect on the dynamic model.

iii - For a link j such that $r_1 \le j < r_2$ and the axis of link j is parallel to that of r_1, the parameters XX_j, XY_j, XZ_j, YY_j, YZ_j, MZ_j have no effect on the dynamic model. The elimination of MX_j,MY_j,M_j must be studied on a case by case basis.

The use of the foregoing results on a tree structure is carried out by decomposing the structure into separate chains beginning from the base.

3-4 Conditions of regrouping the inertial parameters

The regrouping of an inertial parameter X_i, to some other parameters $X_{i1},\dots,X_{ir}$ can be carried out if :

$$DE_i = t_{i1}\, DE_{i1} + \dots \quad + t_{ir}\, DE_{ir} = \sum_{k=1}^{r} t_{ik}\, DE_{ik} \tag{14-a}$$

$$dU_i = t_{i1}\, dU_{i1} + \dots \quad + t_{ir}\, dU_{ir} = \sum_{k=1}^{r} t_{ik}\, dU_{ik} \tag{14-b}$$

with t_{ik} = constant. In fact under these conditions we get the same value of E and U, (and consequently Γ), if we use the parameters X_i, $X_{i1},\dots,X_{ir}$, or if we put X_i equal to zero and the parameters X_{ik} equal to XR_{ik} where:

$$XR_{ik} = X_{ik} + t_{ik} X_i \quad \text{for } k=1,\dots,r \tag{15}$$

3-4-1 General regrouping relations of the inertial parameters

Let the vectors $\mathbf{DE^j}$ and $\mathbf{DU^j}$ be defined as:

$$\mathbf{DE^j} = \left[DE^j_1\ DE^j_2\ \dots\ DE^j_{10}\right]^T;\mathbf{DU^j} = \left[DU^j_1\ DU^j_2\ \dots\ DU^j_{10}\right]^T \tag{16}$$

where:

$DE^j_k = \dfrac{\partial E}{\partial X^j_k}$ and $DU^j_k = \dfrac{\partial U}{\partial X^j_k}$, while X^j_k is the k^{th} element of the vector $\mathbf{X^j}$ (defining the k^{th} inertial parameter of link j).

Calculating $\mathbf{DE^j}$ and $\mathbf{DU^j}$ as function of $\mathbf{DE^i}$ and $\mathbf{DU^i}$; where $a(j) = i$, respectively gives:

$$DE^j_k = \sum_{r=1}^{10} \lambda^j_{k,r}\, DE^i_r + \dot{q}_j f^j_k\ (\mathbf{q}\ ,\ \dot{\mathbf{q}}) \tag{17}$$

$$DU^j_k = \sum_{r=1}^{10} \beta^j_{k,r}\, DU^i_r \tag{18}$$

As $DU^i_r = 0$ for $r = 1,\dots,6$ and taking into account that :

$\beta^j_{k,r} = \lambda^j_{k,r}$ ($r = 7 ,\dots, 10$) [21,22], then (18) can be written as:

$$DU^j_k = \sum_{r=7}^{10} \lambda^j_{k,r}\, DU^i_r = \sum_{r=1}^{10} \lambda^j_{k,r}\, DU^i_r \tag{19}$$

The derivation of λ and f is not given here owing to the limited number of pages, it can be found in references [21, 22].

From the conditions (14) and using the relations (17) and (19), we deduce the following general results:

a- If joint j is rotational then

i- The coefficients $\lambda^j_{9,r}$ and $\lambda^j_{10,r}$ are constant, for $r=1,\dots,10$, while the elements $f^j_9(\mathbf{q}, \dot{\mathbf{q}}) = 0$ and $f^j_{10}(\mathbf{q}, \dot{\mathbf{q}}) = 0$, i.e DE^j_9 and DE^j_{10} verify the condition (14-a). Taking into account the relation (19) , then DU^j_9 and DU^j_{10} verify also the condition (14-b). Thus the parameters M_j and MZ_j , can be regrouped to the parameters of link i.

ii- The sum of DE^j_1 and DE^j_4, corresponding to the elements XX_j and YY_j respectively, can be expressed through constant coefficients as function of DE^i_r (with $r=1,\dots,10$). Thus XX_j , (or YY_j), can be regrouped to YY_j , (or XX_j) and to the parameters of link i We choose to regroup YY_j.

The general regrouping relations corresponding to the elimination of (YY_j,MZ_j,M_j) are given as:

$$\begin{aligned}
XXR_j &= XX_j - YY_j \\
XXR_i &= XX_i + YY_j(1-S\gamma_jS\alpha_j) + 2MZ_j(P_zC\alpha_j-P_yC\gamma_jS\alpha_j) \\
&\quad + M_j(P_y{}^2+P_z{}^2) \\
XYR_i &= XY_i + YY_j\,(CS\gamma_jS\alpha_j) + MZ_j\,(P_x\,C\gamma_jS\alpha_j-P_yS\gamma_jS\alpha_j) \\
&\quad + M_j\,(-P_xP_y) \\
XZR_i &= XZ_i - YY_j(S\gamma_jC\alpha_j) + MZ_j\,(-P_x\,C\alpha_j-P_zS\gamma_jS\alpha_j) \\
&\quad + M_j\,(-P_xP_z) \\
YYR_i &= YY_i + YY_j(1-CC\gamma_jS\alpha_j\,) + 2MZ_j(P_x\,S\gamma_jS\alpha_j+P_zC\alpha_j) \\
&\quad + M_j(P_x{}^2+P_z{}^2) \\
YZR_i &= YZi + YY_j(C\gamma_jCS\alpha_j\,) + MZ_j(-P_y\,C\alpha_j+P_zC\gamma_jS\alpha_j) \\
&\quad + M_j\,(-P_yP_z)
\end{aligned}$$

$$ZZR_i = ZZ_i + YY_jSS\alpha_j + 2MZ_j(-P_x S\gamma_jS\alpha_j - P_yC\gamma_jS\alpha_j) + M_j(P_x^2 + P_y^2)$$

$$MXR_i = MX_i + MZ_j(S\gamma_jS\alpha_j) + M_jP_x$$

$$MYR_i = MY_i - MZ_j(C\gamma_jS\alpha_j) + M_jP_y$$

$$MZR_i = MZ_i + MZ_j C\alpha_j + M_jP_z$$

$$MR_i = M_i + M_j \qquad (20)$$

where: $CC(.) = Cos^2(.)$, $SS(.) = Sin^2(.)$, $CS(.) = Cos(.)Sin(.)$, while P_x , P_y and P_z are the components of ${}^i\mathbf{P}_j$, obtained using (2) and (3) as:

$$ {}^i\mathbf{P}_j = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} d_jC\gamma_j + r_jS\gamma_jS\alpha_j \\ d_jS\gamma_j - r_jC\gamma_jS\alpha_j \\ r_jC\alpha_j + \varepsilon_j \end{bmatrix} \qquad (21)$$

b- if joint j is translational, then :

$\lambda^j_{k,r}$ are constant and $f^j_k(\mathbf{q}, \dot{\mathbf{q}}) = 0$, for $k = 1,\ldots,6$ and $r=1,\ldots,10$. Thus the inertial parameters $1,\ldots,6$ of link j can be regrouped to the inertial parameters of link i. This means that all the parameters of ${}^j\mathbf{J}_j$ will be combined to the parameters of ${}^i\mathbf{J}_i$. The regrouping relation can be obtained as:

$$ {}^i\mathbf{JR}_i = {}^i\mathbf{J}_i + {}^i\mathbf{A}_j\,{}^j\mathbf{J}_j\,{}^j\mathbf{A}_i \qquad (22)$$

3-4-2 Particular regrouping of the inertial parameters

As we have seen in section (3-3) some of the inertial parameters of the first links may have no effect on the dynamic model, so particular regrouping, other than those denoted in section (3-4-1) may take place. The detection of these particular cases have to be studied on a case by case basis using relations (14). We find that the particular regrouping will concern only the parameters MX, MY, MZ and M of the translational links lying between the articulations r_1 and r_2 defined in section (3-3).

3-5 Practical use of the proposed algorithm

The proposed algorithm of the detection of the set of minimum parameters will be carried out as follows:

a- Find the parameters having no effect on the dynamic model, as given in section (3-3)

b-Apply the regrouping relations (20) and (22) from link n to link 1.

c- Apply the relations (14) to the parameters of links $=1,\ldots,r_2$, to detect the existence of supplementary regrouping.

3-6 Example

The given method can be applied to any tree structure robot whatever the values of the geometric parameters or of the number of degrees of freedom, but owing to the limited number of pages and in order to illustrate the method we give the minimum inertial parameters of the three degree of freedom robot of Figure 4. The geometric parameters of the robot are given in Table 1.

Step a : The parameters having no effect on the dynamic model are:

XX_1, XY_1, XZ_1, YY_1 ,YZ_1, MX_1, MY_1, MZ_1 , M_1, MZ_2, and M_2

Step b : The use of relation (20) for $j = 3,\ldots,1$ gives :

link 3:

$XXR3 = XX3 - YY3$
$XXR2 = XX2 + YY3$
$XZR2 = XZ2 - D3\,MZ3$
$YYR2 = YY2 + YY3 + D3^2\,M3$
$ZZR2 = ZZ2 + D3^2\,M3$
$MXR2 = MX2 + D3\,M3$
$MZR2 = MZ2 + MZ3$
$MR2 = M2 + M3$

The minimum parameters of link 3 are :
XXR3, XY3, XZ3, YZ3 , ZZ3 , MX3, MY3 .

link 2:

$XXR2 = XXR2 - YYR2 = XX2 - YY2 - D3^2\,M3$
$ZZR1 = ZZ1 + YY2 + YY3 + D3^2\,M3$

The minimum parameters of link 2 are :
XXR2, XY2, XZR2, YZ2 , ZZR2 , MXR2, MY2 .

link 1 :
The only parameter of first link is ZZR_1.

Step c: The robot having only rotational joints, this step gives no more information .

i	σ	α	d	q	r
1	0	0	0	q_1	0
2	0	90	0	q_2	0
3	0	0	D3	q_3	0

Table 1 : The geometric parameters of the 3 d.o.f Robot

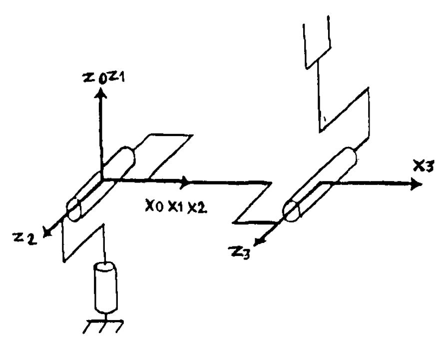

Figure 4: The 3 d.o.f Robot

4- Identification of the inertial parameters

The set of minimum number of inertial parameters are the only identifiable parameters, and they are only needed to the calculation of the control law. They represent, at the case of the given example, 15 parameters, i.e. 15 parameters less than the classical parameters. This will facilitate the identification process. The regrouped relations may not be important in this case, only the knowledge of the parameters to be eliminated, either by regrouping or because they have no effect, is essential. The identification process would give the values of the identifiable parameters directly. In the litterature the identification process is based on the dynamic model as that defined by equation (12), the problem of this model is its need to calculate the joint accelerations. In this section we propose an algorithm for the identification of the inertial parameters and friction coefficients of robots . The algorithm does not require to mesure or to calculate the joint accelerations. The identification model is based on the energy theorem, it is linear in the robot parameters and is easy to calculate.

4-1 Identification model

The energy theorem states that the work of the forces which are applied to a system and which are not derived of a potential is equal to the change of the total energy of the system, thus:

$$\int_{t_1}^{t_2} \dot{\mathbf{q}}^T \mathbf{T}\ dt = [E(t_2) + U(t_2)] - [E(t_1) + U(t_1)]$$
$$= H(t_2) - H(t_1) \tag{23}$$

where :

$\mathbf{T}$ is the torques or forces vector not deriving of a potential,

$E(t_i)$ is the kinetic energy at time t_i,

$U(t_i)$ is the potential energy at time t_i,

$H(t_i)$ is the total energy at time t_i, given as:

$$H(t_i) = E(t_i) + U(t_i) \tag{24}$$

$\mathbf{T}$ can be obtained as :

$$\mathbf{T} = \Gamma + \Gamma_{fs} + \Gamma_{fv} \tag{25}$$

where:

Γ is the motor torques or forces vector,

Γ_{fs} and Γ_{fv} are the coulomb and viscous friction forces vector respectively.

$$\Gamma_{fsj} = -FS_j \operatorname{sign}(\dot{q}_j) \tag{26}$$

$$\Gamma_{fvj} = -FV_j \dot{q}_j \tag{27}$$

where sign(.) denotes the sign function.

As E and U are linear in the inertial parameters then we can write

$$H(t_2) - H(t_1) = \mathbf{DH}^T \mathbf{X} \tag{28}$$

where :

$$\mathbf{DH}^T = [DH_1(t_2) - DH_1(t_1) \ \dots \ DH_m(t_2) - DH_m(t_1)] \tag{29}$$

and

$$DH_i(t_k) = DE_i(t_k) + DU_i(t_k) \tag{30}$$

Let the friction coefficients be defined as :

$$\mathbf{FS} = [FS_1 \dots FS_n]^T \tag{31}$$

$$\mathbf{FV} = [FV_1 \dots FV_n]^T \tag{32}$$

let :

$$\mathbf{DFS}^T = \left[\int_{t_1}^{t_2} |\dot{q}_1|\, dt \ \dots \ \int_{t_1}^{t_2} |\dot{q}_n|\, dt\right] \tag{33}$$

$$\mathbf{DFV}^T = \left[\int_{t_1}^{t_2} \dot{q}_1^2\, dt \ \dots \ \int_{t_1}^{t_2} \dot{q}_n^2\, dt\right] \tag{34}$$

Equation (23) can be rewritten as :

$$y = \mathbf{DH}^T \mathbf{X} + \mathbf{DFS}^T \mathbf{FS} + \mathbf{DFV}^T \mathbf{FV} \tag{35}$$

where:

$$y = \int_{t_1}^{t_2} \dot{\mathbf{q}}^T \Gamma\, dt \tag{36}$$

The vector $\mathbf{X}$ denotes the minimum inertial parameters of the links.

Let us define the vector of the inertial and friction parameters as :

$$\mathbf{K}^T = [\mathbf{X}^T \quad \mathbf{FS}^T \quad \mathbf{FV}^T] \tag{37}$$

so:

$$y = \mathbf{d}^T \mathbf{K} \tag{38}$$

where:

$$\mathbf{d}^T = [\mathbf{DH}^T \quad \mathbf{DFS}^T \quad \mathbf{DFV}^T] \tag{39}$$

Relation (38) represents a linear equation in the inertial and friction parameters of the links, it is function of the joint positions, velocities and of the input joint torques. Most of the robots are provided by position and velocity sensors, and the joint torques constitute the input signal to the motors which are known.

To identify $\mathbf{K}$ a sufficient number of equations can be obtained by calculating the relation (38) between different intervals of times.

The least squares solution is generally used to minimize the mean square error estimation.

4-2 Practical calculation of the functions DU and DE

Assuming that:

$${}^j\omega_j = [\omega_{1,j}\ \omega_{2,j}\ \omega_{3,j}]^T,\ {}^j V_j = [V_{1,j}\ V_{2,j}\ V_{3,j}]^T \tag{40}$$

From the kinetic energy relations (Eqs. 6, 7) and from definition (16), we obtain that :

$$DE_1^j = \frac{1}{2}\omega_{1,j}^2,\ DE_2^j = \omega_{1,j}\omega_{2,j},\ DE_3^j = \omega_{1,j}\omega_{3,j},\ DE_4^j = \frac{1}{2}\omega_{2,j}^2,$$

$$DE_5^j = \omega_{2,j}\omega_{3,j},\ DE_6^j = \frac{1}{2}\omega_{3,j}^2,$$

$$DE_7^j = \omega_{3,j} V_{2,j} - \omega_{2,j} V_{3,j},\ DE_8^j = \omega_{1,j} V_{3,j} - \omega_{3,j} V_{1,j},$$

$$DE_9^j = \omega_{2,j} V_{1,j} - \omega_{1,j} V_{2,j},\ DE_{10}^j = \frac{1}{2}\mathbf{V}_j^T \mathbf{V}_j$$

$$DU_i^j = 0,\ i=1,\dots,6,\ [DU_7^j\ DU_8^j\ DU_9^j] = -\mathbf{g}^T\, {}^0\mathbf{A}_j,$$

$$DU_{10}^j = -\mathbf{g}^T\, {}^0\mathbf{P}_j$$

The components of the velocity vectors can be obtained using equations (8,9)

4-3 Example.

The expressions of DH_i , calculated by a customized method, for the identifiable parameters of the 3 d.o.f robot presented in section (3-6), will be given in this section.

Supposing: $\mathbf{g}^T = [\,0\ \ 0\ \ g\,]$, $Si=\operatorname{Sin}(q_i)$, $Ci=\operatorname{Cos}(q_i)$, we get:

$$DH_6^1 = 1/2\, \dot{q}_1^2$$

$$DH_1^2 = 1/2\, S2^2\, \dot{q}_1^2$$

$$DH_2^2 = S2\, C2\, \dot{q}_1^2$$

$$DH_3^2 = S2\, \dot{q}_1\, \dot{q}_2$$

$$DH_5^2 = C2\, \dot{q}_1\, \dot{q}_2$$

$DH_6^2 = 1/2\, \dot{q}_2^2$

$DH_7^2 = -g\, S2$

$DH_8^2 = -g\, C2$

$\omega_{1,3} = C3\, S2\, \dot{q}_1 + S3\, C2\, \dot{q}_1$

$V_{1,3} = S3\, \dot{q}_2\, D3$

$\omega_{2,3} = -S3\, S2\, \dot{q}_1 + C3\, C2\, \dot{q}_1$

$V_{2,3} = C3\, \dot{q}_2\, D3$

$\omega_{3,3} = \dot{q}_2 + \dot{q}_3$

$V_{3,3} = -C2\, \dot{q}_1\, D3$

$DH_1^3 = 1/2\, \omega_{1,3}^2$

$DH_2^3 = \omega_{1,3}\, \omega_{2,3}$

$DH_3^3 = \omega_{1,3}\, \omega_{3,3}$

$DH_5^3 = \omega_{2,3}\, \omega_{3,3}$

$DH_6^3 = 1/2\, \omega_{3,3}^2$

$DH_7^3 = V_{2,3}\, \omega_{3,3} - V_{3,3}\, \omega_{2,3} + S3\, DH_8^2 + C3\, DH_7^2$

$DH_8^3 = V_{3,3}\, \omega_{1,3} - V_{1,3}\, \omega_{3,3} + C3\, DH_8^2 - S3\, DH_7^2$

5-Conclusion

This paper presents a direct method to determine the set of minimum number of inertial parameters of tree structure robots. The given method permits to determine most of the regrouped parameters by means of closed form relations function of the geometric parameters of the robot. The method is general whatever the values of the geometric parameters. The paper presents also an algorithm for the identification of the inertial parameters and friction coefficients of robots. The algorithm is computationally simple, and it does not require not to mesure or to calculate the joint accelerations. The identification model is based on the energy theorem providing a model linear in the links parameters. These two algorithms are generated automatically by our software package of automatic symbolic modeling of robots SYMORO [23].

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